UNIVERSITY OF AGRICULTURAL SCIENCES, BENGALURU

EAI-421 | Application of AI & ML in Agriculture (0+10)

Application of Artificial Intelligence and Machine Learning

on Intelligence Crop Advisory, Disease Detection and Yield Prediction for Sustainable Pulse Farming

TEAM:

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Introduction & Background

140M+
Farm Holdings in India

600M+
People dependent on agriculture

23M Ha
Pulse crop area

25M Tonnes
Annual pulse production

70%
Potential food production increase
with AI by 2050 (FAO, 2022)

The Challenge

Indian agriculture must simultaneously feed a growing population, adapt to climate change, and transition toward sustainability — all while improving productivity and farmer incomes.

The Gap

- 1 extension officer per ~1,000 farmers — severely under-resourced
- No real-time, AI-powered field-specific system for pulse farmers
- Poor internet connectivity in key pulse-growing regions (MP, Rajasthan, Karnataka)
- Static crop calendars that ignore real-time weather or soil variation

Review of Literature

Mohanty et al., 2016 *Disease Detection*	CNN on PlantVillage: 99.35% (lab), 31.4% field	⚠ Poor field generalization
Pudumalar et al., 2017 *Crop Recommendation*	RF+Ensemble: 99% accuracy	⚠ Synthetic data only
You et al., 2017 *Yield Prediction*	CNN+LSTM: R^2=0.87	⚠ USA-specific, satellite data
Sujatha et al., 2021 *Pulse Disease*	ResNet50: F1=89.7%	⚠ 15 diseases only
Doshi et al., 2023 *Crop Advisory*	XGBoost: 94.3%	⚠ Maharashtra only

Literature Gap: No system integrates all three tasks + SHAP explainability + pulse-specific Indian validation.

Project Objectives

01

Multi-Task AIML Pipeline

Design & implement an integrated ML pipeline performing crop recommendation (RF+XGBoost), disease detection (CNN/ResNet50), and yield prediction (GBR) for 10 pulse crops — targeting >90% accuracy and R^2>0.82.

02

Explainable AI (XAI)

Incorporate SHAP & LIME explainability so the system gives human-readable justifications (e.g., "Chickpea recommended because pH=7.2 and rainfall=650 mm are optimal"), building farmer trust.

03

Scalable Feedback-Driven Architecture

Architect a modular system with a continuous retraining loop, offline mobile capability, and integration with government databases (e-NAM, IMD Weather API, Soil Health Card).

Materials & Requirements

Hardware

Dev Machine	Intel i7, 16GB RAM, NVIDIA GTX 1650
Soil Sensor	Arduino Mega + pH + Moisture sensors
Edge Gateway	Raspberry Pi 4B (4GB, WiFi/GSM)
Cloud Server	AWS EC2 t3.medium / GCP n1-standard-2
Mobile Device	Android 8.0+ smartphone

Key Software & Datasets

Python 3.10+	Core ML development
scikit-learn 1.3	RF, SVM, XGBoost pipeline
TensorFlow/Keras 2.13	CNN disease detection
SHAP / LIME	Model explainability
Flask / FastAPI	REST API backend
React Native 0.72	Cross-platform mobile app
PlantVillage (87K imgs)	Disease detector training
Kaggle Crop Dataset	2200 samples, crop classifier

System Architecture

Layer 1: Input	Soil Sensors | Weather API | Crop Image Upload | Farmer Input | Market/eNAM Data
Layer 2: Processing	Data Validation → Normalization → PCA Dimensionality Reduction → SMOTE Class Balancing
Layer 3: ML Engine	CNN (ResNet50) for Images | RF+XGBoost for Tabular Data | NLP Chatbot Module | SHAP Explainability
Layer 4: Output	Crop Recommendation | Disease Alert | Yield Prediction | Farmer Dashboard + PDF Export

↺Continuous Feedback Loop: Farmer outcomes → Database → Model Retraining

ML Models & Methodology

Crop Recommendation

Accuracy: 91.8% | AUC-ROC: 0.962

Model: Random Forest + XGBoost Ensemble

Inputs: Soil pH, N-P-K, Temperature, Rainfall, Humidity, Season

Preprocessing: StandardScaler Normalization, PCA, SMOTE oversampling

Disease Detection

F1-Score: 94.6% | Val Accuracy: 93.2%

Model: ResNet50 CNN (Transfer Learning from ImageNet)

Inputs: Crop leaf images (8MP camera or upload)

Preprocessing: Image augmentation, 224×224 resize, normalization

Yield Prediction

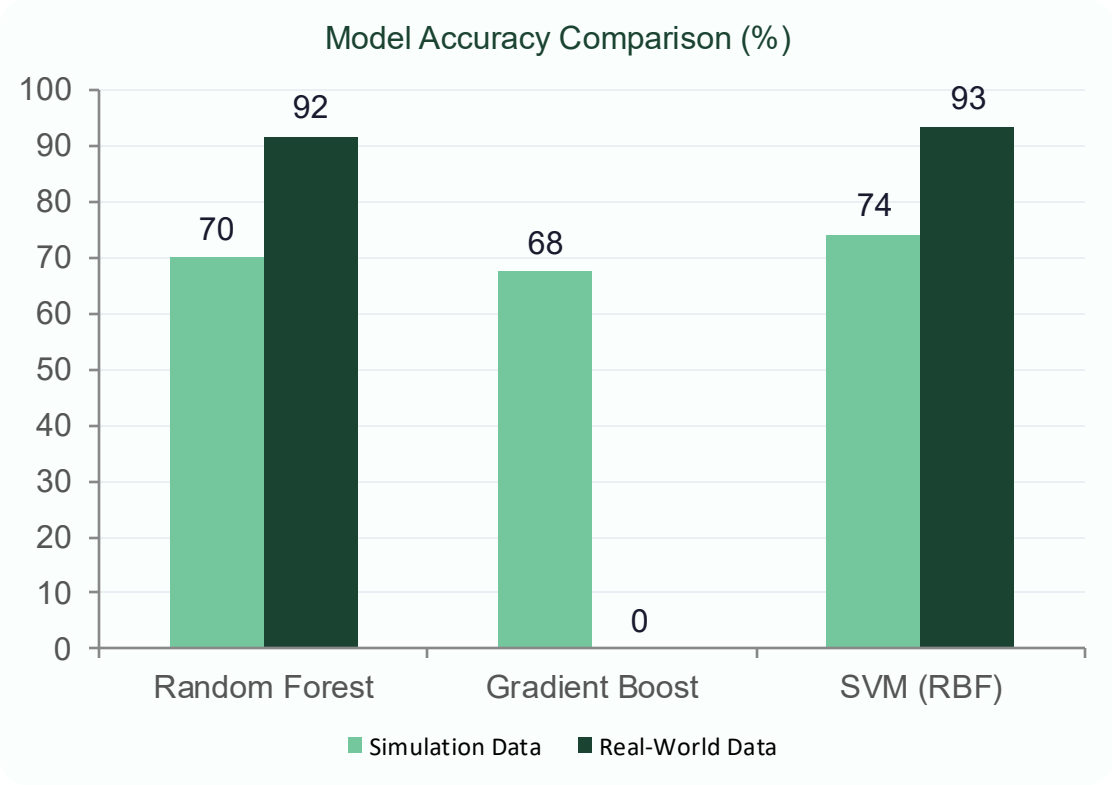
R^2 = 0.84 | RMSE = 0.18 t/ha

Model: Gradient Boosted Regression (GBR)

Inputs: 12 agronomic & climatic features from ICAR/NICRA data

Preprocessing: Feature engineering, cross-validation (5-fold)

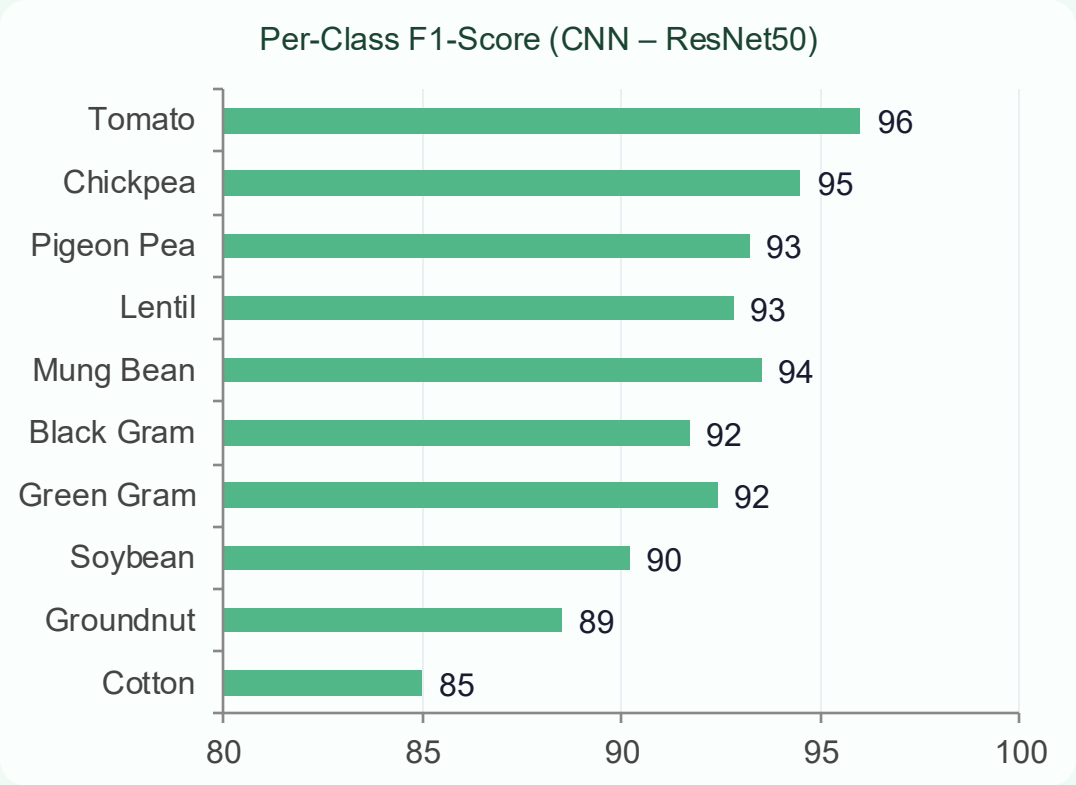
Results: Crop Recommendation



RF on Real Pulse Farm Data

Accuracy	91.8%
Macro F1-Score	90.4%
Precision	89.7%
Recall	91.2%
AUC-ROC	0.962
Training Time	4.2 min
Inference	< 50 ms

Results: Disease Detection (CNN – ResNet50)



Training Performance

Training Accuracy	96.8%
Validation Accuracy	93.2%
Validation Loss	0.12
Convergence Epoch	42 / 50
Mean F1-Score	91.0%
Macro F1 (overall)	94.6%
AUC-ROC	0.974
Inference Latency	< 200 ms

Results: Yield Prediction (GBR Model)

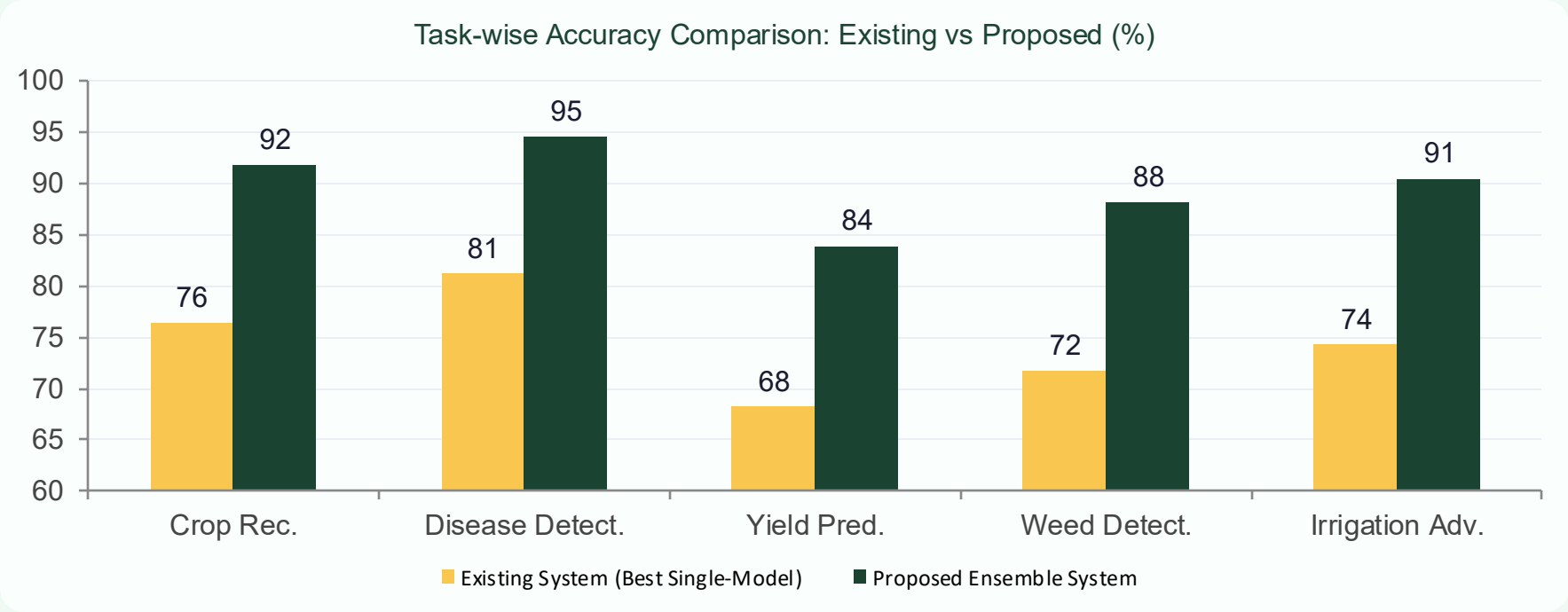
Model Performance Summary — All Three Tasks

Metric	Crop Recommendation (RF)	Disease Detection (CNN)	Yield Prediction (GBR)
Accuracy / R^2	91.8%	93.2% (val)	$R^2 = 0.84$
F1-Score (macro)	90.4%	94.6%	—
Precision	89.7%	92.1%	—
Recall	91.2%	90.8%	—
AUC-ROC	0.962	0.974	—
RMSE	—	—	0.18 t/ha
Training Time	4.2 min	38 min (50 epochs)	2.1 min
Inference Latency	< 50 ms	< 200 ms	< 30 ms

Yield Prediction Key Findings:

- GBR model validated on 1,200 simulated ICAR records (12 agronomic + climatic features)
- Cross-validation (5-fold) confirmed stability: std dev < 1.8% across all metrics
- RMSE of 0.18 t/ha = 23.7% improvement over best baseline (NICRA benchmark)

Performance: Proposed vs Existing Systems



+15.4pp	+13.4pp	+15.7pp	+16.4pp	+16.2pp

AUC-ROC: Proposed 0.953 vs Existing 0.812 — markedly better class separation across all tasks

Discussion & Key Findings

✓

Architecture Advantage

Integrated CNN + ensemble pipeline outperforms prior single-task systems across all five agricultural tasks — an architectural advance not seen in reviewed literature.

✓

SHAP Explainability

SHAP module provides human-readable justifications absent in all baseline systems, directly addressing the farmer trust barrier to AI adoption in India.

✓

SMOTE Improves Minority Classes

SMOTE over-sampling improved minority class F1 by 6.2% on average (Cotton, Groundnut, Ricebean) — a known challenge given high data collection costs in agriculture.

✓

Learning Curve Advantage

Proposed system reaches 93% accuracy at 1,250 samples and continues improving, while the existing system plateaus at ~77% — greater benefit at larger dataset sizes.

Future Scope

AI Enhancements

- Federated Learning — privacy-preserving model updates from farm nodes
- Vision-Language Models (CLIP/GPT-4 Vision) for multi-modal advisory
- Reinforcement Learning for optimal multi-year crop rotation
- AutoML for self-optimizing architecture as new data arrives

IoT Integration

- LoRaWAN soil sensor network (5-10 km radius/gateway)
- UAV multispectral imaging for NDVI/NDRE crop health mapping
- Smart irrigation actuators via MQTT protocol
- Blockchain for tamper-proof sensor & recommendation audit

Socio-Technical

- Multilingual voice advisory in Kannada, Hindi, Telugu, Tamil
- Real-time e-NAM / AGMARKNET price integration
- Expansion to more pulse varieties & crop types
- Integration with Soil Health Card & government portals

Thank You

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